

Polynomial Voxels on an Octree

Placing the Cells where the Medium Needs them, with the Error Known before the Solve

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Abstract

A volume-integral model of seismic waves in a realistically complicated medium cannot afford a uniform grid: real media vary on localised scales, and a uniform grid spends most of its cells where nothing varies. Adaptive trees of polynomial cells, with fast solvers, are established for scalar scattering problems; they are refined to a tolerance, and what the tolerance means for the error of the scattered field is not stated. We ask whether, for elastic waves, cubic voxels on an octree can be placed where the medium needs them with the error of the scattering calculation known before it is run. Each leaf of the tree carries the medium's contrast to polynomial degree r and the wavefield to degree p , and couples to every other leaf through blocks that follow exactly from those between equal cells. We find that the error of the scattered far field is the sum of two terms, both available before the solve. The first is the error of the term of first order in the contrast, which needs no solve: each leaf holds the projection of the incident wave, and for a leaf of uniform contrast its error is a closed-form factor in spherical Bessel functions, with leading term $-(\mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}) d^2/12$ for a constant field in a cell of side d . The second is the nonlinear fraction of the response times the relative projection error of the medium, times a factor of order one that we derive in closed form: it is one in a layer, and in three dimensions it depends on the background and the direction of scattering through the axes of the cells; the nonlinear fraction itself follows from two products with the matrix of a coarse tree, to 1.4%, with no solve. On a sphere with a thin graded shell, against its exact solution, the second term has one constant (0.015) for constant and for linear cells, on uniform grids and on trees, where the raw ratio of error to projection error varies by a factor of two. Refining by both terms, a tree of linear cells reaches the accuracy of a uniform grid with a quarter of the cells on a body whose variation fills 3% of its volume. Which basis suits the medium depends on the medium. Where it varies smoothly the two degrees must be matched, and a constant contrast with a richer field is worse than a constant field. Where it is piecewise constant on the leaves, a constant contrast is exact, and a linear field on the tree has 65 times less error than the uniform grid of constant cells with the same number of unknowns. This draft records what is established and what is not.

1 Introduction

The problem this paper serves. The aim of this series is an efficient model of the seismic wavefield in a medium of realistic complication. Its method is the volume integral equation of elastic scattering [1], discretised into cubic voxels that interact through the Green’s tensor of a background medium, the multiple-scattering description of Foldy [2] and Lax [3]. The first paper of the series [4] determined what a voxel must carry. A voxel represents two functions, the medium within it and the wavefield within it, each in a basis of its own. With the wavefield held to Legendre degree p and the contrast of the medium to degree r , the scheme converges at order $\min(2p + 2, 2r + 2)$ in the cell size, and the error that remains belongs to the single voxel: at second order in the contrast it is the *relative projection error* of the medium, the fraction of the squared norm of its profile that polynomials of degree $\min(p, r)$ on the cells cannot represent. Nestor [4] builds those voxels in three dimensions, in its appendix on the three-dimensional voxel solver.

The question. Those results are for uniform grids. A real medium is not uniform in its difficulty: its variation is concentrated at interfaces, inclusions and thin zones, and a uniform grid fine enough for those is far finer than the rest of the medium needs. The natural cells are then the leaves of an octree, cubes whose sizes differ by powers of two. This paper asks two things. Can the cells be placed where the medium needs them? And can the error of the calculation be known before it is run, from the medium, the tree and the wave alone? The second matters as much as the first. A tree is built by a rule, the rule needs an indicator of where the error is, and an indicator that needs the solution is too late.

What exists. The tree is not new. For the scalar Lippmann–Schwinger equation in two dimensions, Ambikasaran et al. [5] discretise on an adaptive quad tree with a polynomial on every leaf, splitting a leaf until it holds a set number of points per wavelength and resolves the contrast to a tolerance, and Gopal and Martinsson [6] accelerate the direct solve. In three dimensions Malhotra and Biros [7, 8] represent volume sources by polynomials of bounded total degree on the leaves of an adaptive octree, refined where the highest coefficients exceed a tolerance, and evaluate their potentials with a kernel-independent fast multipole method [9] for the Poisson, Stokes and low-frequency Helmholtz problems, at up to 10^{10} unknowns. Polynomials on the cells of a dyadic tree are the multiwavelets of Alpert [10], in which integral operators are sparse [11]. In electromagnetics, large volume elements carry fields of high polynomial order [12], with the material represented to an order of its own [13], and voxels with a linear field are solved with fast Fourier transforms [14]. Where error is estimated for such methods it is estimated *a posteriori*, from a computed solution that is then refined and solved again [15]. For elastic waves the volume integral equation is solved on uniform grids at low order [16], and fast multipole methods exist for boundary integral equations [17, 18].

What this paper adds. Not the tree, the adaptivity or a fast solver. It adds the statement of the error before the solve, for the elastic problem: the two terms of §3, the closed-form factor

of a leaf, and the rule for choosing the basis of the medium. The law is a sharpened instance of a standard fact, that the error of a Galerkin solution of a second-kind equation is governed by the best approximation of the solution; here it is an equality with a factor known in closed form, for the part of the field that copies the medium.

The answer in brief. The error of the scattered field separates by its order in the contrast into two terms, and both are available before the solve (§3). The term of first order is computed in one pass over the leaves, and for a leaf of uniform contrast it is a closed form (Figure 1b). The rest is the nonlinear fraction of the response times the relative projection error of the medium, with one constant for every grid and tree tried, and times a factor of order one that is derived in closed form (§3.4). A refinement rule built on the two terms produces trees that a rule built on the medium alone cannot (§4). And the choice of basis for the medium follows from the medium (§5): a constant in each leaf is the right basis where the medium is piecewise constant on the leaves, and the wrong one where it varies smoothly.

2 Setting

Leaves. The body is covered by cubes, the leaves of an octree: leaf ℓ has centre $\mathbf{x}_\ell$ and half-width h_ℓ (side $d_\ell = 2h_\ell$), and the half-widths are in ratios that are powers of two. With $\xi = (\mathbf{x} - \mathbf{x}_\ell)/h_\ell$ the local coordinate, the leaf carries the nine-component state (displacement and strain) in the Legendre products $Q_a(\xi) = P_{a_1}(\xi_1)P_{a_2}(\xi_2)P_{a_3}(\xi_3)$ of total degree $|a| = a_1 + a_2 + a_3 \leq p_\ell$, which is 1, 4 or 10 functions for $p_\ell = 0, 1$ or 2, and the contrast in the same products to degree r_ℓ . The two degrees may differ from leaf to leaf. A leaf with $p = r = 0$ is a Haar cell: a constant medium and a constant field.

The system. Tested with the same functions, the volume integral equation becomes

$$M_\ell \psi_\ell - \sum_m \mathbf{K}(\ell, m) E_m \psi_m = \langle Q, \psi^0 \rangle_\ell, \quad (1)$$

with ψ_ℓ the coefficients of the state in leaf ℓ , M_ℓ the Gram matrix of its functions, E_m the projected contrast of leaf m acting on its field, ψ^0 the incident wave and $\mathbf{K}(\ell, m)$ the block that couples the two leaves: the Green's tensor integrated against the functions of both [4].

Blocks between leaves of different sizes. A polynomial on a cube is a polynomial of the same degree on each of the cube's descendants, so the block between a large leaf and a small one is an exact, finite sum of blocks between equal cells: the large leaf is replaced by its descendants at the size of the small one, with its polynomials re-expanded (Figure 1c). No integral is needed beyond those between equal cells. This is the two-scale relation of the multiwavelet basis [10]. Against six-dimensional quadrature these blocks agree to 10^{-9} for leaves one and two levels apart, and on a tree whose leaves are all equal the solver reproduces the uniform solver to 10^{-11} .

Symmetry. Two pairs of leaves whose relative positions are images of one another under a reflection or a permutation of the axes have blocks related by the representation of that operation on the polynomials and on the nine components, whatever the two sizes, because the operation acts on both cubes alike. One block is therefore computed for each orbit of the cube’s group of 48 operations. On a tree of 344 leaves of three sizes this reduced the assembly from 125 s to 65 s with an unchanged solution, and made trees with four sizes practicable (§6).

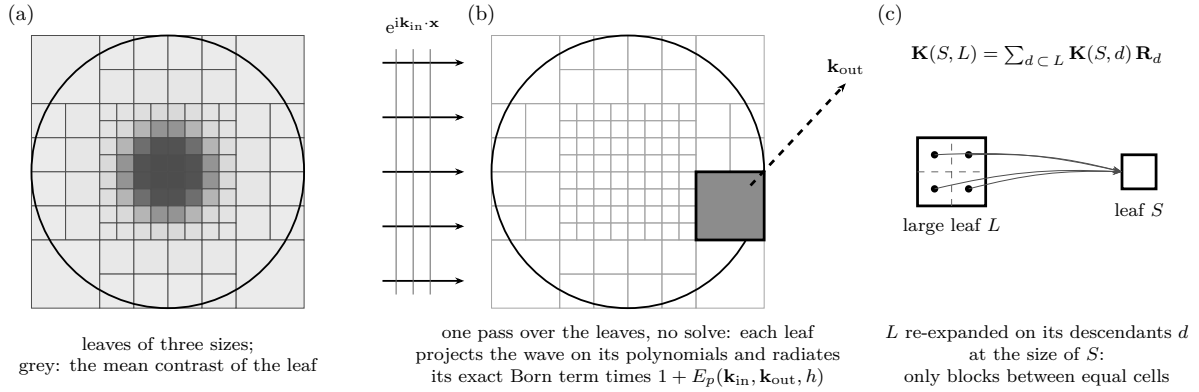


Figure 1. The octree, drawn as its two-dimensional slice. (a) The leaves on a body with a weak halo and a compact graded feature, refined where the feature’s contrast varies, in three sizes; the grey level is the leaf’s mean contrast, which the leaf holds as a polynomial of degree r . (b) The term of first order needs no solve: each leaf holds the projection of the incident wave onto its own polynomials and radiates its exact Born field times the factor $1 + E_p$ of Eq. (2), which depends only on the leaf’s half-width and the two wavevectors (§3.1–§3.2), so the whole term is one pass over the leaves. (c) The block between a large leaf L and a small leaf S is the sum of the blocks between S and the descendants d of L at the size of S , with the polynomials of L re-expanded on them ($\mathbf{R}_d$): only blocks between equal cells are ever integrated. The tree is generated by `figures/make_fig_octree.py`.

3 The error in two terms

Write the scattered far field as a series in the contrast: $u = T_1 + T_2 + \dots$, with T_1 the term of first order (the Born term) and the rest the nonlinear part of the response. The error of the scheme separates the same way.

3.1 The first-order error needs no solve

At first order in the contrast Eq. (1) is $M_\ell \psi_\ell = \langle Q, \psi^0 \rangle_\ell$: each leaf holds the L^2 projection of the incident wave onto its field functions. The scheme’s Born term is therefore known from one pass over the leaves, with no system to solve, and its difference from the Born term of a finer representation (each leaf’s eight children with quadratic cells) is the first-order error of the tree, leaf by leaf. On a tree of two leaf sizes this projected Born term equals the weak-contrast limit of the full solver to 10^{-5} of its size.

3.2 The wave factor of a leaf

For a leaf whose contrast is uniform the first-order error is a closed form. The leaf represents the incident wave $e^{i\mathbf{k}_{\text{in}} \cdot \mathbf{x}}$ by its projection, and it radiates towards the observer through the moments of $e^{-i\mathbf{k}_{\text{out}} \cdot \mathbf{x}}$, with $\mathbf{k}_{\text{out}}$ the P or the S wavevector in the direction of observation. Since $\int_{-1}^1 P_n(t) e^{iqt} dt = 2i^n j_n(q)$, with j_n the spherical Bessel function, the leaf's Born far field is its exact Born far field times

$$1 + E_p = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}. \quad (2)$$

The sum over all degrees is the denominator (the addition theorem), so $E_p \rightarrow 0$ as p grows. Its leading term is

$$E_p = -h^{2p+2} \sum_{|a|=p+1} \prod_{i=1}^3 \frac{(2a_i + 1) (k_{\text{in},i} k_{\text{out},i})^{a_i}}{[(2a_i + 1)!!]^2} + O(h^{2p+4}), \quad E_0 = -\frac{(\mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}) d^2}{12} + \dots \quad (3)$$

For a constant field the error of a leaf therefore depends on the angle of scattering, vanishes at right angles to the incident wave, and differs between the P and the S far field. Along one axis, with $\mathbf{k}_{\text{out}} = \pm \mathbf{k}_{\text{in}}$ of magnitude k , Eq. (3) is $-(\pm 1)^{p+1} c_p (kd)^{2p+2}$ with $c_p = (2p+3)/(4^{p+1}[(2p+3)!!]^2)$: the constants $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ of the layer [4]. The layer's closed-form error is the one-dimensional case of the leaf's.

Checks. Equation (2) is derived symbolically, built a second way without Bessel functions (project, multiply, integrate over the cube) and found identical for $p = 0, 1, 2$; Eq. (3) and its one-dimensional reduction are verified symbolically; and the implementation used below agrees with 30-digit values of the factor to 10^{-13} in 36 cases. Against the solver, on a homogeneous cube (which every grid of cubes represents exactly) at $k_s A = 1$ for half-side A : the Born far field divided by $1 + E_p$ agrees across $p = 0, 1, 2$ and two grids to 9×10^{-7} , where the uncorrected fields differ by up to 14%; and the Born far field of a tree of 15 leaves of mixed sizes and degrees equals the sum over its leaves of exact contribution times factor to 2×10^{-9} .

3.3 The rest is the projection error of the medium

Let $\mathcal{E}_r$ be the relative projection error of the medium on the tree,

$$\mathcal{E}_r = \frac{\sum_{\ell} \int_{\ell} (f - \Pi_r f)^2 dV}{\int f^2 dV}, \quad (4)$$

with $f(\mathbf{x})$ the profile of the contrast and Π_r the L^2 projection onto the polynomials of degree at most r on each leaf. Subtract from the error of the far field the first-order error of §3.1. What is left, divided by $\mathcal{E}_r$, is one constant (Table 1). The test body is a sphere of radius a with a

uniform core of radius $0.75a$ and a shell across which the contrast falls smoothly to zero, at $k_S a = 0.5$, whose exact solution is known; the error is the largest difference of the far field from the exact one over nine scattering angles, relative to its peak. For constant cells the ratio of

Table 1. The error of the far field in two terms, on the sphere with a graded shell. The rest is the error less the first-order error, which is computed without a solve. $\mathcal{E}$ is the relative projection error of Eq. (4) at the degree of the cells. The nonlinear fraction of the exact response is 0.0151.

cells	grid	leaves	error	error / $\mathcal{E}$	rest / $\mathcal{E}$
constant ($p = r = 0$)	uniform	64	8.17×10^{-3}	0.0309	0.0149
	uniform	184	3.18×10^{-3}	0.0235	0.0151
	uniform	408	1.61×10^{-3}	0.0196	0.0154
	uniform	1256	6.77×10^{-4}	0.0166	0.0158
	tree	232	3.29×10^{-3}	0.0321	0.0152
	tree	856	9.28×10^{-4}	0.0210	0.0152
	tree	1360	6.03×10^{-4}	0.0210	0.0151
	tree	1696	5.75×10^{-4}	0.0222	0.0151
linear ($p = r = 1$)	uniform	64	8.99×10^{-4}	0.0147	0.0148
	uniform	184	4.62×10^{-4}	0.0146	0.0146
	uniform	408	2.20×10^{-4}	0.0146	0.0146
	tree	184	5.16×10^{-4}	0.0142	0.0148
	tree	352	2.20×10^{-4}	0.0146	0.0145

error to projection error varies by a factor of 1.93 across the grids; the rest over the projection error varies by 1.06, and it is 0.0151 to 0.0152 on all four trees. For linear cells the first-order error is small and both ratios are near 0.0146. The constant is close to the nonlinear fraction of the exact response, $\nu = |u - T_1|/|u| = 0.0151$. This is the law of the layer [4] carried to three dimensions: the error of the nonlinear part is that part times the relative projection error of the medium, times a factor of order one that §3.4 derives. It holds for constant cells as it does for linear ones, once the first-order error is accounted for; without that, constant cells appear not to follow it.

3.4 The factor of the second term in three dimensions

Why a factor appears. In the layer the relative error of the term of second order in the contrast, T_2 , equals the relative projection error exactly in the long-wave limit, because there the kernel that maps a strain source to the strain it scatters is local: the once-scattered strain is the profile itself, and a cell holds it as the profile’s projection [4]. In three dimensions the same kernel has, besides its local part, a part that falls as the inverse cube of the distance. The once-scattered strain then depends on how the residual of the projection is arranged in space, and the cells of an octree arrange it along their axes. The question is what replaces the layer’s equality.

The second-order term of the scheme. Take a stiffness contrast $f(\mathbf{x}) \Delta \mathbf{C}$, with f the profile, struck by a uniform strain ϵ_{in} and observed through a uniform strain ϵ_{out} of the outgoing P or S wave (the long-wave limit, in which the derivation is made; the test below is at $k_S a = 0.125$).

The second term is the quadratic form $T_2 = \langle f, Kf \rangle$ of the operator K whose Fourier multiplier is

$$m(\hat{\xi}) = \mathbf{a} : \mathbf{\Gamma}(\hat{\xi}) : \mathbf{b}, \quad \mathbf{a} = \Delta \mathbf{C} : \varepsilon_{\text{out}}, \quad \mathbf{b} = \Delta \mathbf{C} : \varepsilon_{\text{in}}, \quad (5)$$

with $\mathbf{\Gamma}$ the static strain Green operator of the background, which depends only on the direction $\hat{\xi}$ of the wavevector: $m = \hat{\xi} \cdot \frac{1}{2}(\mathbf{ab} + \mathbf{ba}) \cdot \hat{\xi} / \mu - \frac{\lambda + \mu}{\mu(\lambda + 2\mu)} (\hat{\xi} \cdot \mathbf{a} \cdot \hat{\xi})(\hat{\xi} \cdot \mathbf{b} \cdot \hat{\xi})$. Cells whose field and contrast are of the same degree compute $\langle \Pi f, K \Pi f \rangle$, Π the projection of §3.3, so with $g = f - \Pi f$ the residual,

$$\frac{T_2 - T_2^{\text{scheme}}}{T_2} = \mathcal{E} F, \quad F = \frac{2\langle g, Kf \rangle - \langle g, Kg \rangle}{\langle f, Kf \rangle \mathcal{E}}. \quad (6)$$

In the layer $F = 1$. For a body whose spectrum is the same in every direction, as a radial profile's is, $\langle f, Kf \rangle = \bar{m} \|f\|^2$ with $\bar{m} = \text{tr}(\frac{1}{2}(\mathbf{ab} + \mathbf{ba})) / (3\mu) - \frac{\lambda + \mu}{\mu(\lambda + 2\mu)} (\text{tr} \mathbf{a} \text{tr} \mathbf{b} + 2 \mathbf{a} : \mathbf{b}) / 15$, the average of m over directions. The two remaining forms are evaluated as the cells shrink, from what the residual looks like inside one cell.

Constant cells. In a cell of half-width h the residual is $\nabla f \cdot (\mathbf{x} - \mathbf{x}_\ell)$, a sum of three sawtooths, one along each axis. Two facts settle the two forms. Its first moments and its energy carry the same constant, $h^2/3$, and its second moments vanish, so its transform at small wavenumbers is $(h^2/3)|\xi|^2 \hat{f}$, the same in every direction, and $2\langle g, Kf \rangle \rightarrow 2\bar{m} \|g\|^2$. And a sawtooth has no mean and its harmonics lie on its own axis, carrying all its energy, so $\langle g, Kg \rangle \rightarrow m_{\text{ax}} \|g\|^2$, with m_{ax} the mean of m over the three cube axes. Hence

$$F_0 = 2 - \frac{m_{\text{ax}}}{\bar{m}}. \quad (7)$$

The cell enters through its axes: a residual arranged along them sees the operator along them, while the body sees it averaged over all directions.

Linear cells. The residual is the quadratic part of f less its projection, $\frac{1}{2} \sum_i H_{ii}(x_i^2 - h^2/3) + \sum_{i < j} H_{ij} x_i x_j$ with H the second derivatives of f . Its energy is $\sum_i H_{ii}^2 h^4 / 45 + \sum_{i < j} H_{ij}^2 h^4 / 9$ per unit volume, and its second moments, $2h^4/45$ on the diagonal and $h^4/9$ off it, give its transform at small wavenumbers as $h^4 W(\xi) \hat{f}$ with $W = \sum_k \xi_k^4 / 45 + \sum_{k < l} \xi_k^2 \xi_l^2 / 9$, which is no longer the same in every direction. The parabolas $x_i^2 - h^2/3$ have their spectrum on their axes; each product $x_i x_j$ of two sawtooths has its spectrum on the lattice of the (i, j) plane, at the points (m, n) , $m, n \neq 0$, with weights $1/(m^2 n^2)$, over which the averages of $u^2 v^2$ and u^4 (u, v the direction cosines) are $6G/\pi^2 - \frac{2}{5}$ and $\frac{9}{10} - 6G/\pi^2$, G being Catalan's constant, from the lattice sum $\sum_{m, n \neq 0} (m^2 + n^2)^{-2} = 4\zeta(2)\beta(2) - 4\zeta(4)$. For a radial profile

$$F_1 = \frac{2\langle Wm \rangle - \frac{1}{225} \sum_i m(\mathbf{e}_i) - \frac{1}{135} \sum_{i < j} M_{ij}}{\frac{8}{225} \bar{m}}, \quad (8)$$

with $\langle Wm \rangle$ the average of Wm over directions and M_{ij} the average of m over the lattice of the (i, j) plane. A constant m gives $F_0 = F_1 = 1$, and so does a layer.

Closed forms. For an incident P wave and an outgoing SV wave the factors depend on the background alone, not on the contrast or the angle:

$$F_0^{\text{PS}} = \frac{6\lambda + 11\mu}{3\lambda + 8\mu}, \quad F_1^{\text{PS}} = \frac{5[(57\lambda + 85\mu)\pi^2 - 315G(\lambda + \mu)]}{28\pi^2(3\lambda + 8\mu)}, \quad (9)$$

1.516 and 1.337 for the background of the tests ($\alpha = 5000$, $\beta = 3000 \text{ ms}^{-1}$). For an outgoing P wave the factors are ratios of expressions linear in $\cos 2\theta$, θ the scattering angle; for the contrast of the tests F_0^{PP} runs from 1.101 near forward and back scattering to 0.923 at right angles, and F_1^{PP} from 1.066 to 0.950.

The test. The derivation is symbolic, in 31 checks, from the Navier equation to the closed forms (Mathematica/AdaptiveOctree_T2Factor.wl). It is tested two ways on a sphere of radius $a = 10 \text{ m}$ whose contrast is uniform within $a/2$ and falls smoothly to zero at the surface, with a stiffness contrast, at $k_s a = 0.125$: by the scheme itself, whose T_2 is set against the exact one, angle by angle and wave by wave (scripts/measure_t2_law_3d.py); and by evaluating the two forms of Eq. (6) directly, by fast Fourier transform on a grid finer than the cells (scripts/measure_t2_quadratic_forms.py); the details are in Appendix A. On 8 cells the

Table 2. The factor F of Eq. (6): the scheme on 8 cells across, the direct evaluation on finer grids (32 cells for constant, 24 for linear), and the limit, Eqs. (7)–(9).

outgoing wave	constant cells			linear cells		
	scheme	direct	limit	scheme	direct	limit
P at 0.2 rad	1.074	1.092	1.101	1.073	1.067	1.066
P at $\pi/2$	0.944	0.930	0.923	0.948	0.949	0.950
SV at 0.2 rad	1.376	1.470	1.516	1.356	1.342	1.337

scheme and the direct evaluation agree to 1% where both are computed on the same grid, which confirms Eq. (6). As the cells shrink both approach the limit: for constant cells $\langle g, Kg \rangle / (\bar{m} \|g\|^2)$ goes 0.926, 0.909, 0.902 on 8, 16 and 32 cells against its limit $m_{\text{ax}}/\bar{m} = 0.899$ (outgoing P at 0.2 rad), slowly, the distance falling as about $h^{1.7}$; for linear cells the factor is within 0.4% of the limit by 24 cells. The scheme on 8 cells is three quarters of the way to the limit for constant cells.

What it means for the second term. The second term of the error is the nonlinear part of the response times the relative projection error times F , a factor of order one that depends on the background, the direction of scattering and the type of wave, and on the cells through their axes, not on the grid. For this background and contrast it lies between 0.92 and 1.52 for constant cells and between 0.95 and 1.34 for linear cells, and it is one in a layer, which is why the layer's law is an equality. The constant of Table 1, formed from the largest error over the angles and with every order above the first included, is 0.96 to 1.05 times ν , inside that range. The second term is therefore known before the solve to within F , and in closed form once the direction of observation is chosen. The derivation holds for the term of second order, in the long-wave limit, for a body whose spectrum is isotropic; the terms of higher order are measured, not derived.

3.5 The nonlinear fraction without a solve

Why it is needed. The second term of the error is $\nu \mathcal{E} F$, and of its three factors only the nonlinear fraction $\nu = \max|u - T_1| / \max|u|$ (the maxima over the angles of observation) has so far needed a solve: one solve on a coarse tree that resolves the medium, taking $|u - T_1|/|u|$ of that tree. That estimate is low, by up to a quarter for constant cells (Table 3), and the reason is the second term itself: the coarse tree’s nonlinear part is low by its own relative projection error.

The method. The scheme’s response is a series in the contrast, $\psi = \sum_n \psi_n$ with $M\psi_1 = b$ and $M\psi_{n+1} = KE\psi_n$, M the Gram matrices of the leaves, b the moments of the incident wave and KE the coupling times the contrast (Eq. (1)). The first term is the Born term of §3.1; each further term costs one product with the assembled matrix and no solve. Two facts make a few terms enough. On the exact solutions the terms of second and third order carry the nonlinear fraction to 0.2% (on the three bodies of Table 3; T_2 alone to 4%), because $|T_3|/|T_2|$ is 0.03 to 0.04. And on the coarse tree the nonlinear part of the scheme is low by its relative projection error $\mathcal{E}$, which is known before the solve (§3.3). The estimate is therefore

$$\nu \approx \frac{\max|T_2 + T_3|/(1 - \mathcal{E})}{\max|T_1 + (T_2 + T_3)/(1 - \mathcal{E})|}, \quad (10)$$

with T_n the far fields of the scheme’s terms on the coarse tree and the factor F of §3.4 taken as one.

The result. Table 3 sets the estimates against the exact nonlinear fraction on the three bodies of the paper, for constant and linear cells, the coarse tree being that of the refinement experiments (the rule on the medium alone at a tolerance of 10^{-2} from a tree of eight leaves; `scripts/measure_octree_nu_estimate.py`; Appendix A). The estimate of Eq. (10) is within

Table 3. The nonlinear fraction estimated on a coarse tree, over its exact value. “Solve” is the earlier estimate (one dense solve); Eq. (10) uses two products with the matrix and the tree’s projection error $\mathcal{E}$.

body	cells	leaves	$\mathcal{E}$	exact ν	solve	$T_2/(1 - \mathcal{E})$	Eq. (10)
thin shell, $k_S a = 0.5$	constant	64	0.264	0.0151	0.746	1.047	1.014
	linear	64	0.061		0.940	1.039	1.000
feature $0.2a$ – $0.5a$, $k_S a = 1$	constant	120	0.175	0.0107	0.816	1.017	0.990
	linear	120	0.036		0.963	1.028	0.999
feature $0.1a$ – $0.3a$, $k_S a = 1$	constant	176	0.145	0.0059	0.854	1.025	1.000
	linear	176	0.015		0.985	1.027	0.999

1.4% of the exact value in all six cases; the solve is 25% low at worst, and the second-order term alone, corrected, is 2% to 5% high, which is the share of the third-order term.

What it means. Both terms of the error are now known before the solve, without any solve: the first from one pass over the leaves, the second from the relative projection error, the factor F in closed form, and the nonlinear fraction from two products with the matrix of a coarse tree.

The estimate rests on the law of §3.3 applied to the coarse tree, and has been tested on three bodies at two frequencies.

3.6 The terms of third and higher order

Why they matter. The law of §3.3 was measured on the whole nonlinear part, $u - T_1$, and §3.4 derived the factor of its second-order term only. Each further term carries an error of its own, and in the layer the whole nonlinear part does not follow the second-order law: all orders together give 0.90 to 0.94 times $\mathcal{E}$ where T_2 alone gives one [4].

The layer, exactly. In the long-wave limit the layer's kernel is local, the strain scattered by a stiffness contrast being $c f \varepsilon$ with $c = -\Delta M / M_p$, $M_p = \lambda + 2\mu$. With field and contrast of the same degree in each cell the scheme's term of order n is then $c^{n-1} \int (\Pi f)^n dz$ against the exact $c^{n-1} \int f^n dz$, for $n \leq 3$ at every degree and for every n in constant cells, so the relative error of each term is a property of the profile,

$$e_n = \frac{\int f^n dz - \int (\Pi f)^n dz}{\int f^n dz}, \quad e_2 = \mathcal{E}. \quad (11)$$

Summed, the exact strain is $\varepsilon_0 / (1 - cf)$, and the scheme's is the Galerkin solution of $\varepsilon = \varepsilon_0 + c f \varepsilon$ in each cell separately, on that cell's polynomials: the error of the whole nonlinear part is the single site's, in closed form. On the layer of the companion paper (profile $1 + \frac{1}{2} \sin(2\pi z/D)$) at $\omega = 30 \text{ rad s}^{-1}$ the scheme's T_3 has the error e_3 of Eq. (11) to 10^{-4} , for constant, linear and quadratic cells on 4 to 16 cells, in reflection and in transmission, and to 0.5% at $\omega = 300 \text{ rad s}^{-1}$; $e_3/\mathcal{E} = 2.455$ on every grid, which is $3 \langle f \rangle_{g^2} \int f^2 dz / \int f^3 dz$ with $\langle f \rangle_{g^2}$ the mean of the profile weighted by the squared residual. Because $T_3/T_2 = -0.078$, the second and third terms together have 0.877 times $\mathcal{E}$, and the whole nonlinear part at the full contrast 0.896 times $\mathcal{E}$, which the cell-by-cell Galerkin solution reproduces to four digits with $c = -0.0640$ from the contrast; the exact layer's T_3/T_2 returns the same c to five digits (`scripts/measure_layer_t3_low.py`). This is the companion paper's 0.90.

Three dimensions. The kernel is not local, and the third-order term involves the once-scattered strain fields themselves: with $\mathbf{u}_b = \mathbf{\Gamma} * (f\mathbf{b})$ and $\mathbf{u}_a = \mathbf{\Gamma} * (f\mathbf{a})$ in the notation of Eq. (5), $T_3 = \int f \mathbf{u}_a : \Delta \mathbf{C} : \mathbf{u}_b dV$, and constant cells compute $\int \Pi f \Pi \mathbf{U}_a : \Delta \mathbf{C} : \Pi \mathbf{U}_b dV$ with $\mathbf{U} = \mathbf{\Gamma} * (\Pi f \cdot)$. Its factor F_3 , defined as F is by Eq. (6), therefore depends on the body's own field and not only on averages of m .

The closed form. The difference is expanded to second order in the half-width, as for the second term, with $\mathbf{u} = \Pi \mathbf{u} + \mathbf{u}'$ the cell decomposition of each field. Four pieces appear, each from a lemma already in hand. The residual g meets a smooth field $w_{ab} = \mathbf{a} : \mathbf{\Gamma} * (f \Delta \mathbf{C} : \mathbf{u}_b)$ through its first moments, giving $(h^2/3) \nabla f \cdot \nabla w_{ab}$. The field scattered by the residual is a sawtooth on each cube axis carrying $\mathbf{\Gamma}$ on that axis, $\mathbf{G}_{ib} = \mathbf{\Gamma}(\mathbf{e}_i)\mathbf{b}$, with amplitude $\partial_i f$. The smooth fields vary across a cell as sawtooths of amplitude $\partial_i \mathbf{u}$. And a cell's mean of a product of three fields splits

exactly into the product of the means and the means of products of the departures. The terms in $(\partial_i f)^2 \mathbf{G}\mathbf{G}$ cancel, and

$$\begin{aligned} \frac{T_3 - T_3^{\text{scheme}}}{h^2/3} = \int & \left[\nabla f \cdot \nabla (w_{\mathbf{a}\mathbf{b}} + w_{\mathbf{b}\mathbf{a}}) \right. \\ & + f \sum_i (\partial_i \mathbf{u}_{\mathbf{a}} : \Delta \mathbf{C} : \partial_i \mathbf{u}_{\mathbf{b}} - \partial_i f (\partial_i \mathbf{u}_{\mathbf{a}} : \Delta \mathbf{C} : \mathbf{G}_{i\mathbf{b}} + \mathbf{G}_{i\mathbf{a}} : \Delta \mathbf{C} : \partial_i \mathbf{u}_{\mathbf{b}})) \\ & + \sum_i \partial_i f (\mathbf{u}_{\mathbf{a}} : \Delta \mathbf{C} : \partial_i \mathbf{u}_{\mathbf{b}} + \partial_i \mathbf{u}_{\mathbf{a}} : \Delta \mathbf{C} : \mathbf{u}_{\mathbf{b}}) \\ & \left. - \sum_i (\partial_i f)^2 (\mathbf{u}_{\mathbf{a}} : \Delta \mathbf{C} : \mathbf{G}_{i\mathbf{b}} + \mathbf{G}_{i\mathbf{a}} : \Delta \mathbf{C} : \mathbf{u}_{\mathbf{b}}) \right] dV, \end{aligned} \quad (12)$$

with $F_3 = (T_3 - T_3^{\text{scheme}})/(T_3 \mathcal{E})$ and $\mathcal{E} = (h^2/3) \int |\nabla f|^2 dV / \int f^2 dV$. It is a closed form in the sense of Eq. (7): the cells enter only through $\mathbf{\Gamma}$ on their three axes, and everything else is an integral of the body's smooth, once-scattered static fields, so it needs no grid of cells. In a layer every field is local and the integrand is $3ff'^2 \mathbf{G}_{\mathbf{a}} : \Delta \mathbf{C} : \mathbf{G}_{\mathbf{b}}$, which is the layer's e_3 ; the same expansion of the second term gives Eq. (7). The algebra is verified symbolically in 7 checks (Mathematica/AdaptiveOctree_F3.wl).

The test. Equation (12) is evaluated on the sphere of §3.4 by fast Fourier transform of the smooth fields alone, converged in resolution and in the padding against the images of the periodic box (scripts/measure_f3_closed_form.py; its value of F for the second term is 1.1016 against the exact 1.1009). It is set against two evaluations on cells: the scheme, whose terms are those of its Born series (scripts/measure_t3_law_3d.py), and the cell forms computed directly with $\mathbf{\Gamma}$ applied by fast Fourier transform (scripts/measure_t3_tensor_forms.py). On 4 cells across the last two agree to 0.3% for the outgoing P wave. The details of every measurement are in Appendix A.

Linear cells. With the field and the contrast in $1, x_1, x_2, x_3$ the projection on the outgoing side can be moved onto the profile, and the scheme's term is $\int \Pi f \mathbf{U}_{\mathbf{a}} : \Delta \mathbf{C} : \Pi \mathbf{U}_{\mathbf{b}} dV$, with one projection, on the incident side; for constant cells it is the form above. The residual is now the quadratic part of the profile less its projection, $\frac{1}{2} \sum_i f_{,ii} P_i + \sum_{i<j} f_{,ij} Q_{ij}$ with $P_i = x_i^2 - h^2/3$ and $Q_{ij} = x_i x_j$ (local coordinates), and every term reduces to two cell rules. A cell-scale function meets a smooth one through their second derivatives,

$$B(S, T) = h^4 \left[\sum_i \frac{S_{,ii} T_{,ii}}{45} + \sum_{i<j} \frac{S_{,ij} T_{,ij}}{9} \right], \quad (13)$$

and the field the residual scatters carries $\mathbf{\Gamma}$ on the axes for its parabolas and $\mathbf{M}_{ij}$, the average of $\mathbf{\Gamma}$ over the lattice of the (i, j) plane with $\langle u^2 v^2 \rangle = 6G/\pi^2 - \frac{2}{5}$ (§3.4), for its products. The term in two scattered residual fields is absent, because the projection removes the part of $\mathbf{U}_{\mathbf{b}}$ at the

scale of the cell, and

$$T_3 - T_3^{\text{scheme}} = \int \left[B(f \Delta \mathbf{C} \mathbf{u}_a, \mathbf{u}_b) + B(f, w_{ab} + w_{ba} + \mathbf{u}_a : \Delta \mathbf{C} : \mathbf{u}_b) - H(f \Delta \mathbf{C} \mathbf{u}_a, f; \mathbf{b}) \right. \\ \left. - f K(f; \mathbf{a}, \mathbf{u}_b) - \mathbf{u}_a : \Delta \mathbf{C} : B(f, \mathbf{u}_b) - K_2(f; \mathbf{a}) : \Delta \mathbf{C} : \mathbf{u}_b \right] dV, \quad (14)$$

where H , K and K_2 are B with one smooth factor replaced by the scattered residual,

$$H(S, f; \mathbf{b}) = h^4 \left[\sum_i \frac{S_{,ii} : \mathbf{G}_{ib} f_{,ii}}{45} + \sum_{i < j} \frac{S_{,ij} : \mathbf{M}_{ij} \mathbf{b} f_{,ij}}{9} \right],$$

$$K(f; \mathbf{a}, \mathbf{u}) = h^4 \left[\sum_i \frac{f_{,ii} \mathbf{G}_{ia} : \Delta \mathbf{C} : \mathbf{u}_{,ii}}{45} + \sum_{i < j} \frac{f_{,ij} \mathbf{M}_{ij} \mathbf{a} : \Delta \mathbf{C} : \mathbf{u}_{,ij}}{9} \right],$$

$$K_2(f; \mathbf{a}) = h^4 \left[\sum_i \frac{f_{,ii}^2 \mathbf{G}_{ia}}{45} + \sum_{i < j} \frac{f_{,ij}^2 \mathbf{M}_{ij} \mathbf{a}}{9} \right],$$

and $\mathcal{E} = h^4 \int [\sum_i f_{,ii}^2/45 + \sum_{i < j} f_{,ij}^2/9] dV / \int f^2 dV$. In a layer the integrand is $(h^4/45) \kappa (6f'^2 f'' + 3ff''^2)$, $\kappa = \mathbf{G}_a : \Delta \mathbf{C} : \mathbf{G}_b$, which is the direct expansion of $\int f^3 - \int (\Pi f)^3$; and the same rules give the second term as $2B(f, \mathbf{a} : \mathbf{u}_b) - \mathbf{a} : K_2(f; \mathbf{b})$, whose ratio is the F_1 of Eq. (8). The algebra is verified in 8 checks (`Mathematica/AdaptiveOctree_F3Linear.wl`). The closed form gives $F_3 = 1.475, 1.428$ and 1.387 for the outgoing P wave at $0.2, 0.885$ and $\pi/2$ rad and 1.892 for the SV wave, and its value of F_1 for the second term is 1.0672 against the exact 1.0659 . The direct evaluation on cells approaches it at second order in the cell size, and extrapolated from 16 and 24 cells gives $1.475, 1.436, 1.404$ and 1.874 . The scheme on 8 cells gives $1.350, 1.299, 1.254$ and 1.79 , which the direct evaluation on 8 cells reproduces ($1.351, 1.264$ and 1.79): for linear cells F_3 , unlike F_1 , is still far from its limit on 8 cells.

Table 4. The factors of the second and third terms and of the whole nonlinear part, $k_s a = 0.125$: the scheme on 8 cells across, the direct evaluation of F_3 (constant cells: 32 cells, outgoing P, the SV wave not converged; linear cells: extrapolated from 16 and 24), and the closed forms, Eqs. (12) and (14).

outgoing wave	F (scheme)	F_3 (scheme)	F_3 (direct)	F_3 (closed form)	T_3/T_2	nonlinear part
constant cells						
P at 0.2 rad	1.073	1.600	1.586	1.594	-0.036	1.054
P at $\pi/2$	0.944	1.494	1.417	1.410	-0.042	0.921
SV at 0.2 rad	1.375	2.145		2.46	-0.021	1.358
linear cells						
P at 0.2 rad	1.066	1.350	1.475	1.475	-0.036	1.056
P at $\pi/2$	0.948	1.254	1.404	1.387	-0.042	0.935
SV at 0.2 rad	1.341	1.79	1.874	1.892	-0.021	1.332

The direct evaluation converges as the cells shrink ($1.62, 1.58, 1.59$ on 8, 16 and 32 cells, outgoing P at 0.2 rad) to the closed form, 1.594 , and agrees with it to 0.5% at both angles. The scheme on 8 cells is three quarters of the way to it for the SV wave, as it is for the second term. The nonlinear part's factor is the mean of the terms' factors weighted by the terms: $(F + F_3 T_3/T_2)/(1 + T_3/T_2) = 1.054$ against the measured 1.054 , so the fourth and higher terms contribute 0.05% .

What it means. Every order has its own factor, larger than the second's, and the factor of the nonlinear part is their mean weighted by the terms. Where the contrast is moderate the second term dominates ($|T_3/T_2|$ is 0.02 to 0.04 here) and the nonlinear part's factor is F to within $|T_3/T_2|(F_3 - F)$, two percent here; the estimate of §3.5 already carries the third term. In the layer the whole series is derived; in three dimensions the factor of the third term is derived in closed form for constant and for linear cells, Eqs. (12) and (14), and those of the higher terms are measured.

4 Refinement by the two terms

The rule. Each leaf has two indicators, both computed before any solve. The *medium* indicator is the nonlinear fraction ν times the leaf's share of the projection error, $\nu \int_{\ell} (f - \Pi_r f)^2 dV / \int f^2 dV$. The *wave* indicator is the leaf's own first-order error (§3.1) relative to the peak of the Born far field. A leaf is split into its eight children while the sum of the two exceeds a tolerance. In the experiments below the nonlinear fraction is estimated from one solve on a coarse tree that resolves the medium, as $|u - T_1|/|u|$ of that tree; Eq. (10) now gives it without the solve and more accurately (§3.5). The predicted error of a tree is the first-order error of the whole tree, summed over the leaves as complex amplitudes, plus $\nu \mathcal{E}_r$. It uses no reference solution.

A localised feature. The test body is a sphere of radius a whose contrast is a weak halo, falling smoothly from 5% of the central value to zero across the whole radius, plus a compact feature in which it falls from the central value to the halo between $0.1a$ and $0.3a$: under 3% of the volume. Its exact solution is known. At $k_5 a = 1$, with leaves no smaller than $a/32$, the errors are those of Table 5. The rule that uses the medium alone refines a leaf while its share of the projection error exceeds a tolerance.

Constant cells. The rule that uses the medium alone fails: with 512 leaves its error is twice that of the uniform grid of 408 cells. It refines the feature and leaves the halo coarse, and the halo is where the first-order error is, in leaves whose medium is too smooth for the rule to see. The rule with two terms sees it. Its tree of 260 leaves has the error of the uniform grid of 408, and against the trend of the uniform grids its trees have the same error with about 35% fewer cells at the coarse end and an error 15 to 20% lower at the fine end. The saving is modest, and the reason is the cell: a constant field has a second-order error in every leaf, so the halo needs cells however smooth its medium is.

Linear cells. Here the tree pays. The tree of 176 leaves has eleven times less error than the uniform grid of 184 cells. By the trend of the two uniform grids its error would need about 730 uniform cells, four times as many, and that of the tree of 452 leaves about 1130; these uniform equivalents are extrapolated, not computed. The fourth-order error of a linear field lets the halo stay coarse, and the tree then follows the medium. The two-term rule improves on the medium alone by about 15% at about 340 leaves.

Table 5. The localised feature: leaves, the error predicted without a reference, and the error against the exact solution. The nonlinear fraction estimated from a coarse tree is 0.0051 (constant cells) and 0.0059 (linear cells); the exact value is 0.0059.

cells	grid	leaves	predicted	error
constant	uniform	408	1.11×10^{-2}	7.15×10^{-3}
	uniform	1256	4.36×10^{-3}	2.92×10^{-3}
	medium alone	344	1.60×10^{-2}	1.55×10^{-2}
	medium alone	512	1.57×10^{-2}	1.53×10^{-2}
	medium alone	736	5.92×10^{-3}	5.49×10^{-3}
	two terms	260	8.18×10^{-3}	7.18×10^{-3}
	two terms	372	6.82×10^{-3}	5.38×10^{-3}
	two terms	752	4.48×10^{-3}	3.78×10^{-3}
	two terms	1780	2.18×10^{-3}	1.75×10^{-3}
linear	uniform	184	2.83×10^{-3}	2.73×10^{-3}
	uniform	408	8.27×10^{-4}	6.77×10^{-4}
	medium alone	176	2.43×10^{-4}	2.47×10^{-4}
	medium alone	344	1.95×10^{-4}	1.96×10^{-4}
	two terms	260	1.72×10^{-4}	1.68×10^{-4}
	two terms	336	1.56×10^{-4}	1.66×10^{-4}
	two terms	452	1.08×10^{-4}	1.14×10^{-4}

The prediction. The predicted error is above the true error on all eleven grids of constant cells computed for this body (nine of them in Table 5), by 3% to 55%. For linear cells the ratio of predicted to true is 0.94 to 1.22 on nine grids: close, but not a bound.

5 Which basis for the medium

A smooth medium: the degrees must be matched. On the body of §4, hold the contrast constant in every leaf and raise the field to linear. The error rises, at four times the unknowns: from 7.15×10^{-3} to 8.31×10^{-3} on the uniform grid of 408 cells, and from 7.18×10^{-3} to 8.32×10^{-3} on the tree of 260 leaves. The prediction, made without a solve, says so beforehand (1.11×10^{-2} against 1.34×10^{-2} on the uniform grid). On a smoothly varying medium the first-order error of a leaf with a constant contrast is not the wave’s: it is the error of holding the contrast constant, which a richer field cannot repair, and with a constant field the wave’s error partly cancels it. This is the rule of Nestor [4] that the two bases must be matched.

A medium that is piecewise constant on the leaves. The pairing of a constant medium with a polynomial field belongs to the medium that is constant in each leaf. The test body is a cube of half-side B with a uniform contrast a quarter of the central value, holding at its centre a cube of half-side $B/4$ with the full contrast, at $k_s B = 2$. A tree of 120 leaves represents it exactly with one constant per leaf ($\mathcal{E}_0 = 0$): 56 leaves of half-width $B/4$ and 64 of half-width $B/8$. The uniform grid that does the same has 512 cells. No exact solution is known for this body; the reference is the tree with quadratic fields, whose own first-order error is predicted to be 2×10^{-7} (Table 6). With the medium exact there is no second term, and the first-order prediction is

Table 6. A cube within a cube, the contrast constant in every leaf. The first-order error is predicted without a solve; the error is against the tree with quadratic fields. In the last two rows the faces of the inner cube fall inside the cells of the grid.

grid and field	cells	unknowns	predicted	error
tree, constant field	120	1080	1.54×10^{-2}	1.53×10^{-2}
tree, linear in the 56 large leaves	120	2592	3.24×10^{-3}	3.21×10^{-3}
tree, linear field	120	4320	8.07×10^{-5}	8.42×10^{-5}
uniform, constant field	512	4608	5.54×10^{-3}	5.51×10^{-3}
uniform, linear field	512	18432	8.17×10^{-6}	9.25×10^{-6}
uniform, constant field	64	576	1.81×10^{-2}	1.79×10^{-2}
uniform, linear field	64	2304	3.01×10^{-2}	2.92×10^{-2}

within 13% of the measured error in all seven cases and within 4% in five. The tree with a linear field has an error of 8.4×10^{-5} with 4320 unknowns; the uniform grid of constant cells has 5.5×10^{-3} with 4608: 65 times more. Raising the field to linear in the large leaves alone lowers the tree’s error from 1.5×10^{-2} to 3.2×10^{-3} : a large leaf needs a richer basis for the wavefield, not for the medium. And where the faces of the inner cube fall inside cells, the linear field is worse than the constant one, as on the smooth body; for the scalar problem Anderson et al. [19] likewise find that high order is lost at jumps of the medium unless the grid follows them.

The choice. The two cases give the rule. Where the medium is piecewise constant and the tree can follow its interfaces, hold it constant in each leaf and spend the degree on the field. Where it varies smoothly, give the medium and the field the same degree and let the tree follow the medium. In both the error is predicted before the solve.

6 What is established, and what is not

- *Established twice.* The wave factor, Eqs. (2) and (3): a symbolic derivation and an independent numerical implementation, and both against the solver.
- *Established against exact solutions, by one implementation.* The two-term separation (Table 1) and the refinement results (Table 5): one frequency each, two bodies, both spheres with radial profiles.
- *Self-consistent only.* The cube within a cube (Table 6) has no exact solution; its reference is the same scheme with quadratic fields. The field near the edges and corners of the cubes is not smooth, and an independent solution is needed before the figure of 65 is relied on.
- *Derived, and established twice.* The factor of the second term, Eqs. (6) to (9): a symbolic derivation, and the scheme and a direct evaluation against it (Table 2). It is derived for the term of second order, in the long-wave limit, for a body with an isotropic spectrum.
- *Established against exact solutions, by one implementation.* The nonlinear fraction from two products with the matrix of a coarse tree, Eq. (10): within 1.4% on three bodies, for

constant and linear cells (Table 3). The refinement experiments of §4 were run with the earlier estimate from a coarse solve and have not been repeated with it.

- *Derived in the layer; measured twice in three dimensions.* The terms of third and higher order (§3.6): in the layer each term’s error and the whole series in closed form, the latter the cell-by-cell Galerkin solution of the local problem; in three dimensions the third term’s factor for constant cells in closed form, Eq. (12), verified symbolically and against the scheme and a direct evaluation, and for linear cells, Eq. (14), likewise. The factors of the fourth and higher terms are not derived.
- *Not a bound.* The predicted error exceeded the true error for constant cells in every case, but fell 6% below it for linear cells.
- *Size.* The solver is dense and stops near 20 000 unknowns: about 2200 constant leaves or 550 linear ones. The saving in cells is therefore shown on small bodies. An equal-cell block costs about 50 ms, and a tree of 344 leaves of four sizes is assembled and solved in about three minutes.
- *Earlier work.* The survey behind §1 found no statement of the two-term law, of the leaf’s factor or of the matched-degree rule, and no elastic volume scheme with polynomial cells. It rests on some twenty searches and on the opening pages of most of the works cited; the adaptive solvers of Ambikasaran et al. [5] and Malhotra and Biros [8] have yet to be read in full for any estimate that relates their refinement tolerance to the error of the solution.

7 Conclusions

On an octree of polynomial voxels the error of the scattered field is two terms, and both are known before the solve: the first-order error, from one pass over the leaves and in closed form for a leaf of uniform contrast, and the nonlinear fraction of the response times the relative projection error of the medium, times a factor of order one that is one in a layer and in three dimensions depends, in closed form, on the background and the direction of scattering through the axes of the cells; the nonlinear fraction comes from two products with the matrix of a coarse tree, so neither term needs a solve. The closed form of the first term reduces in one dimension to the constants of the layer. A refinement rule built on both terms places cells where a rule built on the medium alone cannot, and with linear cells a tree reaches the accuracy of a uniform grid with a quarter of the cells on a body whose variation is localised. A constant contrast in each leaf is the right basis for a medium that is piecewise constant on the leaves, where a linear field on the tree outperforms the uniform grid of constant cells sixty-five-fold at equal unknowns in a self-consistent test, and the wrong one for a smooth medium, where the two degrees must be matched. What remains before the method models a realistic medium is a solver whose cost does not grow as the square of the number of leaves. Its parts are published: the fast multipole method for volume potentials on an octree [8], the treatment of the elastic kernels in fast multipole methods for boundary integral equations [17, 18], and, for cells mapped to a uniform grid, the adaptive integral method [20]. What the error analysis gives any of them is

the tolerance they must meet: the first-order error and the projection error are known before the solve, so a fast product need be no more accurate than they are.

The octree, its blocks and its solver are `cubic_scattering/graded_voxel/octree.py`, with tests in `cubic_scattering/tests/test_graded_voxel_octree.py`. The wave factor is derived in `Mathematica/AdaptiveOctree_WaveFactor.wl` and measured by `scripts/measure_octree_wave_term.py`; the factor of the second term is derived in `Mathematica/AdaptiveOctree_T2Factor.wl` and measured by `scripts/measure_t2_law_3d.py` and `scripts/measure_t2_quadratic_forms.py`; Table 3 is `scripts/measure_octree_nu_estimate.py`; the terms of higher order are `scripts/measure_layer_t3_law.py`, `scripts/measure_t3_law_3d.py` and `scripts/measure_t3_tensor_forms.py`, with the closed form of the third term derived in `Mathematica/AdaptiveOctree_F3.wl` and evaluated by `scripts/measure_f3_closed_form.py`; Table 1 is `scripts/measure_octree_two_term.py`; Table 5 is `scripts/pilot_octree_two_term_refinement.py`; the constant medium with a richer field is `scripts/pilot_octree_hp_refinement.py`; Table 6 is `scripts/pilot_octree_blocky_body.py`. Their results are stored in `LatexPDFs/AdaptiveOctree/figures`.

A The experiments in detail

Every measurement of §§3.4–3.6 is set out here: the body, the background and the contrast, the waves and how they are observed, how each term of the series is extracted, what is neglected and how small it is, and the numerical parameters of each evaluation.

Background, contrast and waves. The background is isotropic with $\alpha = 5000$ and $\beta = 3000 \text{ m s}^{-1}$ and $\rho = 2500 \text{ kg m}^{-3}$. The contrast is $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu = 1 \text{ GPa}$ and $\Delta\rho = 100 \text{ kg m}^{-3}$ times the profile, or the stiffness alone ($\Delta\rho = 0$) where the long-wave derivations of §§3.4 and 3.6 are tested, since the density enters the second and third terms only at higher order in $k_s a$. The incident wave is a P wave along axis 1. The scattered field is observed at nine angles from 0.2 to $\pi - 0.2$ rad in the plane of axes 1 and 2, at 5×10^8 radii, where the near-field terms are negligible, and split into its P part (radial) and its S part (transverse).

The bodies. All are spheres of radius $a = 10 \text{ m}$ whose contrast is a function of radius, built from the quintic step $s_5(x) = 10x^3 - 15x^4 + 6x^5$ on $[0, 1]$, so that their exact solutions are known by the radial reference of the voxel papers [4]. The sphere of §§3.4 and 3.6 has full contrast within $a/2$ and $s_5((a-r)/(a/2))$ outside it. The thin-shell sphere of §§3.3 and 3.5 has full contrast within $0.75a$. The two bodies with a feature (§§3.5 and 4) have a halo $0.05 s_5((a-r)/(a-r_1))$ plus $0.95 s_5((r_2-r)/(r_2-r_1))$, with the feature between $r_1 = 0.2a$ and $r_2 = 0.5a$, or between $0.1a$ and $0.3a$.

The terms of the series. The scheme's terms are those of its Born series (`octree.born_series_octree`), each one product with the assembled matrix, on uniform trees of 4, 6 and 8 cells across (§3.4 uses the equivalent uniform solver of the voxel papers, with blocks by the cube's symmetry). The exact terms are read from the exact sphere at scaled contrasts $\pm d$ and $\pm 2d$: $T_1 = [8(u_d - u_{-d}) - (u_{2d} - u_{-2d})]/12d$, $T_2 = [16(u_d + u_{-d}) - (u_{2d} + u_{-2d})]/24d^2$ and $T_3 = [(u_{2d} - u_{-2d}) - 2(u_d - u_{-d})]/12d^3$, u_s the field at contrast s times the full one (§3.4 uses $T_2 = (u_d + u_{-d})/2d^2$ at $d = 10^{-3}$, whose error, of order $d^2 T_4/T_2$, is negligible there). Each carries a truncation error of order d^2 times the ratio of the next term of its parity, and a cancellation: T_3 is recovered from the odd part through a fraction $3d^2 T_3/T_1$ of it, about 10^{-7} at $d = 10^{-2}$. For constant cells, whose $\mathcal{E}$ is about 0.06 on 8 cells, $d = 10^{-2}$ suffices. For linear cells, whose $\mathcal{E}$ is 5×10^{-3} , the error to be measured in T_3 is a few parts in 10^3 , and $d = 10^{-2}$ gave a factor that was not symmetric about $\pi/2$ (which the static derivation requires) and that worsened at lower frequency: the cancellation, not the physics. With $d = 0.05, 0.1$ and 0.2 the factor is symmetric, its truncation falls as d^2 (successive differences in the ratio 4), and the Richardson extrapolations from the two pairs agree to 10^{-4} ; these are the values given. The frequency is $k_s a = 0.125$, where the half-width of a cell is 0.016 wavelengths and the dynamic corrections to the factors are negligible; the derivations are static.

The measured ratio. For each angle and each wave type the ratio is $\text{Re}[(T - T^{\text{scheme}}) \cdot \bar{T}] / (|T|^2 \mathcal{E})$, T the exact term as a displacement vector; for the S wave at $\pi/2$ the second and third terms vanish and the ratio is not defined. $\mathcal{E}$ is the relative projection error of the profile on the cells, by a Gauss rule of 10 points per axis in each cell (§3.4) or of 6 points on descendants three levels deep (§§3.5 and 3.6).

The direct evaluations on cells. The forms of Eq. (6) and of the third term are evaluated by fast Fourier transform with the profile sampled at s midpoints per cell and axis, on a periodic box larger than the body by a padding factor, with $\mathbf{\Gamma}$ at zero wavevector set to its average over directions. The cell projection is the discrete one on the samples, which keeps the residual orthogonal to the cell's functions on them and converges as $1/s$; the exact L^2 projection, sampled, leaves spurious low moments in the residual that are harmless for constant cells and large for the small residual of linear cells. For the second term the exact $\langle f, Kf \rangle = \bar{m} \|f\|^2$ of a radial profile replaces the transform's own value, which carries a small-wavevector error of 1% to 10%. The grids run from $s = 4$ to 32 and 4 to 32 cells across, on boxes of up to 480^3 points; for linear cells $s = 4$ is too coarse and is not used.

The closed forms. Equations (7), (8) and (9) are evaluated exactly (`Mathematica/AdaptiveOctree_T2Factor.wl`). Equations (12) and (14) need the body's smooth static fields, which are computed by fast Fourier transform with 40 to 96 points across the body and a padding of 2 to 5; the factors are converged in resolution to four digits, and the padding, against the periodic images whose coupling through the inverse-cube kernel decays slowly, is checked by the second term, whose closed form is known: at a padding of 5 it is 1.1016 against 1.1009 for the P wave and 1.521 against 1.516 for the SV wave (`scripts/measure_f3_closed_form.py`).

The nonlinear fraction. The coarse tree is the rule on the medium alone at a tolerance of 10^{-2} , from a tree of $2 \times 2 \times 2$ leaves, with leaves no smaller than $a/16$ ($a/32$ for the feature between $0.1a$ and $0.3a$). Its relative projection error $\mathcal{E}$ is that of the profile on its leaves at their degree. The exact nonlinear fraction is formed from the exact field and the exact Born term, the latter by a central difference at $d = 10^{-3}$ (`scripts/measure_octree_nu_estimate.py`).

The layer. The layer of §3.6 is that of the companion paper: thickness $D = 2$ m, the profile $1 + \frac{1}{2} \sin(2\pi z/D)$, a plane source at $z = -12$ m, observers at $z = -1$ m (reflection) and $z = 4$ m (transmission), the contrast above with its density, normal incidence, 4, 8 and 16 cells of degree 0, 1 and 2, at $\omega = 30$ and 300 rad s^{-1} . Its terms are read as above with $d = 10^{-2}$, from the scheme and from the exact layer in 30-digit arithmetic. The constant of the local kernel is $c = -\Delta M/M_p = -0.0640$, with $\Delta M = \Delta\lambda + 2\Delta\mu$ and $M_p = \lambda + 2\mu$, and the cell-by-cell Galerkin solution is computed with the Gauss rule of the cells (`scripts/measure_layer_t3_law.py`).

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